SCOPE OF ACCREDITATION TO ISO/IEC 17025:2005
& ANSI/NCSL Z540-1-1994

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CALIBRATION

Valid To: June 30, 2018 Certificate Number: 1509.01

In recognition of the successful completion of the A2LA evaluation process, accreditation is granted to this laboratory to perform the following calibrations[1]:

I. Dimensional

Parameter/Equipment	Range	CMC[2, 4] (±)	Comments
Calipers[3, 5]	Up to 24 in (24 to 48) in (48 to 96) in	0.6*R* 1100 µin 1500 µin	Gage blocks (field) Gage blocks (lab)
Dial and Test Indicators[3, 5]	Up to 3 in	99 µin	Gage blocks, ULM, SuperMic[10]
Gage Blocks	Up to 1 in (2 to 4) in (4 to 20) in	$(2 + 1.5L)$ µin $(3 + 1.5L)$ µin $(6.7 + 1.9L)$ µin	Master gage block set, Federal 130B-24 comparator Maser gage block set, ULM
Height Gages[3, 5]	Up to 36 in	0.6*R*	Gage blocks
Length Standards	Up to 18 in Up to 48 in	$(37 + 1.5L)$ µin $(65 + 1.5L)$ µin	ULM/SuperMic[10], gage blocks

Parameter/Equipment	Range	CMC[2, 4] (±)	Comments
Levels	Up to 0.0001 in	0.67 DIV	Surface plate, granite square
Micrometers[3, 5]	Up to 4 in (4 to 24) in (24 to 48) in (48 to 96) in	33 µin 0.6*R* 1100 µin 1500 µin	Gage blocks (field) Gage blocks (lab)
Plain Ring Gages	(0.125 to 12) in	(22 + 1.5*L*) µin	ULM, gage blocks
Protractors	0 to 90 °	0.049 °	Surface plate, angle blocks
Thread Plugs – Pitch Diameter Major Diameter Pitch Diameter (Taper)	 (0.040 to 12) in (0.040 to 12) in (0.040 to 12) in	 (69 + 1.5*L*) µin (18 + 1.5*L*) µin (75+ 1.5*L*) µin	 Three-wire method, direct measure Two-wire method, direct measure
Thread Rings Pitch Diameter Taper Thread Rings Pitch Diameter	Up to 3.125 in Up to 3 in	250 µin 120 µin	Setting plug Setting plug, standoff to master
Thread Wires	(4 to 80) TPI	(11 + 1.5*L*) µin	Master thread wire set
Surface Plates[3,5] Indicator Stands	(1.5 X 1.5) to (4 X 8) ft Up to 12 X 12 in Up to 12 X 12 in	31 µin 45 µin 580 µin	Repeat reading gage Repeat reading gage, triangular gage block base Indicator, triangular gage block base

Parameter/Equipment	Range	CMC[2, 4] (±)	Comments
Optical Comparators[3,5] X-Axis Y-Axis Angle	Up to 12 in Up to 12 in (0 to 90) Degrees	(250 + 91L) µin (250 + 91L) µin 0.029 Degrees	Glass scale Angle blocks
Rulers[3,5] Tape Measures[3,5] PI Tapes[3,5]	Up to 40 in Up to 40 in Up to 100 ft Up to 72 in	0.013 in 0.013 in 0.062 in 910 µin	Standard ruler Standard ruler Standard tape Gage blocks
Radius Gages	Up to 0.75 in	1300 µin	Optical comparator and overlay
Angle Gages	Up to 180 Degrees	0.044 Degrees	Optical comparator and vernier
Torque Arms	Up to 24 in Up to 48 in	0.0047 in 0.003 in	Height gage, test indicator Gage blocks

II. Electrical – DC/Low Frequency

Parameter/Range	Frequency	CMC[2, 6, 7, 9] (±)	Comments
AC Current – Measure 1 A 3 A	60 Hz to 1 kHz	0.1 % + 0.4 mA 0.1 % + 1.8 mA	HP 34401

Parameter/Range	Frequency	CMC[2, 6, 7, 9] (±)	Comments
AC Current – Generate			Wavetek 9100
(32 to 320) µA 320 µA to 3.2 mA (3.2 to 32) mA (32 to 320) mA 320 mA to 3.2 A (3.2 to 10) A	60 Hz to 3 kHz	0.072 % + 0.9 µA 0.07 % + 0.6 µA 0.07 % + 3.2 µA 0.083 % + 32 µA 0.1 % + 480 µA 0.21 % + 3 mA	
(3.2 to 32) A	(60 to 100) Hz (100 to 440) Hz	0.2 % + 5.5 mA 0.78 % + 27 mA	using 10 turn coil
(32 to 200) A	(60 to 100) Hz (100 to 440) Hz	0.21 % + 90 mA 0.67 % + 0.25 A	
(16 to 160) A	(60 to 100) Hz	0.21 % + 28 mA	
(160 to 1000) A	(60 to 100) Hz	0.21 % + 0.45 A	using 50 turn coil
AC Voltage – Generate			
(32 to 320) mV 320 mV to 3.2 V (3.2 to 32) V (32 to 105) V	60 Hz to 3 kHz	0.041 % + 20 µV 0.041 % + 200 µV 0.041 % + 2.0 mV 0.043 % + 6.3 mV	Wavetek 9100
(105 to 320) V	60 Hz to 1 kHz (1 to 3) kHz	0.066 % + 20 mV 0.09 % + 20 mV	
(320 to 800) V	(60 to 100) Hz (1 to 3) kHz	0.079 % + 63 mV 0.1 % + 63 mV	
AC Voltage – Measure			
100 mV 1 V 10 V 100 V 750 V	60 Hz to 3 kHz	0.065 % + 40 µV 0.064 % + 300 µV 0.064 % + 3 mV 0.064 % + 30 mV 0.064 % + 230 mV	HP 34401
Up to 10k V		0.14 % + 100 mV	Vitrek 4700

Parameter/Equipment	Range	CMC[2, 6, 7, 9] (±)	Comments
Capacitance – Generate			
Low	(0.5 to 4) nF (4 to 40) nF (40 to 400) nF 400 nF to 4 µF	0.61 % + 15 pF 0.6 % + 60 pF 0.61 % + 320 pF 0.8 % + 3.2 nF	Wavetek 9100
High	(4 to 40) µF (40 to 400) µF 400 µF to 4 mF (4 to 40) mF	1 % + 32 nF 1 % + 320 nF 1 % + 3.2 µF 2 % + 120 µF	
DC Current – Generate	(0 to 320) µA 320 µA to 3.2 mA (3.2 to 32) mA (32 to 320) mA 320 mA to 3.2 A (3.2 to 11) A	0.014 % + 11 nA 0.014 % + 83 nA 0.014 % + 900 nA 0.016 % + 9.6 µA 0.06 % + 120 µA 0.057 % + 940 µA	Wavetek 9100
	(3.2 to 32) A (32 to 100) A (100 to 160) A	0.19 % + 1.2 mA 0.072 % + 9.4 mA 0.068 % + 45 mA	using 10 turn coil
	(160 to 520) A (520 to 1000) A	0.056 % + 47 mA 0.055 % + 230 mA	using 50 turn coil
DC Current – Measure, Fixed Points	10 mA 100 mA 1 A 3 A	0.076 % + 6 µA 0.05 % + 5 µA 0.1 % + 100 µA 0.14 % + 600 µA	HP 34401
DC Voltage – Generate	(0 to 320) mV 320 mV to 3.2 V (3.2 to 32) V (32 to 320) V (320 to 1050) V	0.0063 % + 4.2 µV 0.0062 % + 42 µV 0.0066 % + 420 µV 0.0069 % + 4.5 mV 0.0084 % + 20 mV	Wavetek 9100
DC Voltage – Measure, Fixed Points	100 mV 1 V 10 V 100 V 1000 V	0.0055 % + 3.5 µV 0.0042 % + 7 µV 0.0036 % + 50 µV 0.0047 % + 600 µV 0.0047 % + 10 mV	HP 34401
	Up to 10k V	0.071 % + 30 mV	Vitrek 4700

Parameter/Equipment	Range	CMC[2, 6, 7, 9] (±)	Comments
Electrical Calibration of Temperature Controllers[3, 5]	(-200 to 1371) °C	0.56 °C	Omega CL27
Resistance – Generate	(0 to 40) Ω (40 to 400) Ω (0.4 to 4) kΩ (4 to 40) kΩ (40 to 400) kΩ (0.4 to 4) MΩ (4 to 40) MΩ (40 to 400) MΩ	0.11 % + 50 mΩ 0.051 % + 100 mΩ 0.037 % + 200 mΩ 0.052 % + 2 Ω 0.053 % + 20 Ω 0.049 % + 200 Ω 0.15 % + 2 kΩ 0.14 % + 40 kΩ	Wavetek 9100
Resistance – Measure, Fixed Points	100 Ω 1 kΩ 10 kΩ 100 kΩ 1 MΩ 10 MΩ 100 MΩ	0.01 % + 4 mΩ 0.01 % + 10 mΩ 0.01 % + 100 mΩ 0.01 % + 1 Ω 0.01 % + 10 Ω 0.04 % + 100 Ω 0.8 % + 10 kΩ	HP 34401
RTD Simulation – PT385, 100 Ω	 (-200 to -100) °C (-100 to 100) °C (100 to 630) °C (630 to 850) °C	 0.26 °C 0.17 °C 0.35 °C 0.53 °C	 Wavetek 9100
Thermocouple Simulation – Type E Type J	 (-250 to -200) °C (-200 to -100) °C (-100 to 100) °C (100 to 1000) °C (-210 to -100) °C (-100 to 800) °C (800 to 1000) °C (1000 to 1200) °C	 0.45 °C 0.23 °C 0.18 °C 0.22 °C 0.26 °C 0.20 °C 0.22 °C 0.24 °C	 Wavetek 9100

Parameter/Equipment	Range	CMC[2, 6, 7, 9] (±)	Comments
Thermocouple Simulation – (cont)			
Type K	(-250 to -200) °C (-200 to -100) °C (-100 to 100) °C (100 to 600) °C (600 to 1372) °C	0.57 °C 0.28 °C 0.20 °C 0.24 °C 0.28 °C	Wavetek 9100
Type T	(-250 to -200) °C (-200 to -100) °C (-100 to 0) °C (0 to 400) °C	0.59 °C 0.28 °C 0.23 °C 0.18 °C	
Thermocouple – Measure			
Type E	(-270 to -100) °C (-100 to -25) °C (-25 to 650) °C (650 to 1000) °C	0.51 °C 0.21 °C 0.19 °C 0.23 °C	Martel 3001
Type J	(-210 to -100) °C (-100 to -30) °C (-30 to 760) °C (760 to 1200) °C	0.29 °C 0.23 °C 0.20 °C 0.25 °C	
Type K	(-270 to -100) °C (-100 to -25) °C (-25 to 120) °C (120 to 1000) °C (1000 to 1372) °C	0.34 °C 0.24 °C 0.29 °C 0.28 °C 0.41 °C	
Type T	(-270 to -150) °C (-150 to 0) °C (0 to 400) °C	0.64 °C 0.26 °C 0.19 °C	
Welding Devices	(0 to 350) Amps DC (0 to 100) Volts DC (100 to 700) Feed Rate IPM	0.87 ADC 0.012 VDC 3.6 IPM	Load bank, current shunt, DMM

III. Mechanical

Parameter/Equipment	Range[8]	CMC[2] (±)	Comments
Balances[3, 5]			Handbook 44 with:
	(0 to 20) g	0.11 mg	Class 1 weights
	(0 to 200) g	0.69 mg	
	(0 to 200) g	1.3 mg	Class 2 weights
	(0 to 1000) g	6.6 mg	
	(0 to 5000) g	34 mg	
	(0 to 20) g	0.81 mg	Class 4 weights
	(0 to 200) g	4.8 mg	
	(0 to 1000) g	24 mg	
	(0 to 5000) g	130 mg	
	(0 to 10 000) g	370 mg	
	(0 to 20 000) g	460 mg	
Force – Universal Testing Machines[3, 5] –			
Tension / Compression	(0 to 100) lbf	0.0064 lbf	Class F weights w/ load cells
Compression	(100 to 1000) lbf	0.26 % Indication	
	(1000 to 10 000) lbf	0.24 % Indication	
	(10 000 to 50 000) lbf	0.33 % Indication	
	(10 000 to 100 000) lbf	0.21 % Indication	
	(50 000 to 500 000) lbf	0.21 % Indication	
Tension	(100 to 1000) lbf	0.23 % Indication	
	(1000 to 10 000) lbf	0.23 % Indication	
	(10 000 to 50 000) lbf	0.29 % Indication	
Mass, Fixed Points	1 mg	2.2 µg	Modified double substitution
	2 mg	2.1 µg	
	5 mg	2.9 µg	
	10 mg	1.4 µg	
	20 mg	1.5 µg	
	30 mg	2.2 µg	
	50 mg	1.6 µg	
	100 mg	1.7 µg	
	200 mg	1.6 µg	
	300 mg	1.6 µg	
	500 mg	2.1 µg	
	1 g	11 µg	
	2 g	3.3 µg	
	5 g	5.7 µg	
	10 g	7.2 µg	

Parameter/Equipment	Range[8]	CMC[2] (±)	Comments
Mass (cont)	20 g 50 g 100 g 200 g 500 g 1 kg 2 kg 5 kg	13 µg 11 µg 39 µg 140 µg 320 µg 390 µg 1.7 mg 2.6 mg	Modified double substitution
	10 kg 20 kg	93 mg 94 mg	Direct comparison
	0.8859 g ($^1/_{32}$ oz) 1.772 g ($^1/_{16}$ oz) 3.544 g ($^1/_8$ oz) 7.087 g ($^1/_4$ oz) 14.17 g ($^1/_2$ oz) 28.35 g (1 oz) 56.7 g (2 oz) 113.4 g (4 oz) 226.8 g (8 oz)	12 µg 94 µg 240 µg 170 µg 170 µg 370 µg 150 µg 740 µg 5.1 mg	
	0.4536 g (0.001 lb) 0.9072 g (0.002 lb) 2.27 g (0.005 lb) 4.54 g (0.01 lb) 9.07 g (0.02 lb) 22.68 g (0.05 lb) 45.36 g (0.1 lb) 90.72 g (0.2 lb)	130 µg 56 µg 26 µg 16 µg 56 µg 79 µg 37 µg 87 µg	
	453.6 g (1 lb) 907.2 g (2 lb) 2267.96 g (5 lb) 4535.9 g (10 lb) 9071.8 g (20 lb) 22 679.62 g (50 lb) 45 359.237 g (100 lb) 90 718.475 g (200 lb) 226 796.185 g (500 lb) 453 592.37 g (1000 lb)	3.8 mg 7.9 mg 8.3 mg 7.4 mg 94 mg 95 mg 0.43 g 0.43 g 0.45 g 0.47 g	
Pipettes[3, 5]	(1 to 50) µL (50 to 100) µL (100 to 500) µL (500 to 1000) µL (1000 to 5000) µL (5000 to 10 000) µL	0.31 µL 0.6 µL 1.3 µL 2.2 µL 12 µL 18 µL	Gravimetric method

Parameter/Equipment	Range	CMC² (±)	Comments
Optical Tachometer[3, 5]	200 FPM 3000 FPM 29 999 FPM	1.3 FPM 1.7 FPM 3.9 FPM	Calibrated strobe
Pressure Gages, Transducers and Transmitters[3,5]	(0 to 30000) psig (0 to 10000) psig (0 to 3000) psig (0 to 5000) psig (0 to 300) psig (0 to 100) psig (0 to 500) psig	35 psig 12 psig 3.5 psig 3.9 psig 0.18 psig 0.03 psig 0.19 psig	Additel digital test gauge Crystal XP2 digital test gauge Druck/Martel transducer Transmation 195
Scales[3, 5]	(0 to 10) lb (0 to 20) lb (0 to 50) lb (0 to 100) lb (0 to 200) lb (0 to 500) lb (0 to 1000) lb (0 to 5000) lb (0 to 20 000) lb	0.00025 lb 0.00025 lb 0.0063 lb 0.0063 lb 0.0064 lb 0.0064 lb 0.18 lb 0.19 lb 0.21 lb	Handbook 44 with: Class 4 weights Class F weights
Torque Wrenches[3,5]	(1 to 50) in·lbf (30 to 400) in·lbf (80 to 1000) in·lbf (240 to 3000) in·lbf (1200 to 6000) in·lbf (2400 to 12 000) in·lbf	0.17 in·lbf 1.8 in·lbf 5.1 in·lbf 21 in·lbf 23 in·lbf 58 in·lbf	CDI/Norbar torque calibrator Norbar torque calibrator
Torque Transducers	Up to 200 in·lbf Up to 1200 in·lbf Up to 12 000 in·lbf	0.10 % of rdg 0.091 % of rdg 0.10 % of rdg	Torque arm and weights

Parameter/Equipment	Range	CMC[2] (±)	Comments
Vacuum Gauges[3, 5]	(0 to 15) psig (0 to 15) psig	0.03 psig 0.03 psig	Transmation 195 Druck DPI 610
Rotations Speed Measurement	Contact: 200 RPM 3000 RPM 20 000 RPM Optical: 3000 RPM 30 000 RPM	 1.5 RPM 2.4 RPM 11 RPM 1.3 RPM 3.3 RPM	Monarch PLT200

IV. Thermodynamic

Parameter/Equipment	Range	CMC[2] (±)	Comments
Temperature – Measuring Equipment[3, 5]	(30 to 300) °C (86 to 572) °F	0.09 °C 0.16 °F	Hart 1502A with 5628 PRT in dry block
Temperature – Measuring Equipment	(-40 to 200) °C (-40 to 392) °F	0.061 °C 0.11 °F	Hart 1502A/5628 PRT in wet bath, in lab only
Temperature –Measure[3, 5]	(-100 to 600) °C (-148 to 1112) °F	0.64 °C 1.2 °F	Fluke 51 with thermocouple
IR Thermometer[3,5]	(-20 to 150) °C (Amb +10 to 400) °C	0.33 °C 1.8 °C	Mikron M340 (lab only) Omega BB703
Humidity – Measure[3, 5]	(10 to 90) % RH	3.0 % RH	Newport iTHX

V. Time & Frequency

Parameter/Equipment	Range	CMC[2, 7] (±)	Comments
Frequency – Measure	40 Hz to 300 kHz	0.014 % Indication	HP 34401
Timers and Stopwatches[3, 5]	< 1 hr ≥ 1 hr	0.14 s/24 hr 0.59 s/24 hr	Reference stopwatch

[1] This laboratory offers commercial calibration service and field calibration service.

[2] Calibration and Measurement Capability Uncertainty (CMC) is the smallest uncertainty of measurement that a laboratory can achieve within its scope of accreditation when performing more or less routine calibrations of nearly ideal measurement standards or nearly ideal measuring equipment. CMCs represent expanded uncertainties expressed at approximately the 95 % level of confidence, usually using a coverage factor of $k = 2$. The actual measurement uncertainty of a specific calibration performed by the laboratory may be greater than the CMC due to the behavior of the customer's device and to influences from the circumstances of the specific calibration.

[3] Field calibration service is available for this calibration and this laboratory meets A2LA *R104 – General Requirements: Accreditation of Field Testing and Field Calibration Laboratories* for these calibrations. Please note the actual measurement uncertainties achievable on a customer's site can normally be expected to be larger than the CMC found on the A2LA Scope. Allowance must be made for aspects such as the environment at the place of calibration and for other possible adverse effects such as those caused by transportation of the calibration equipment. The usual allowance for the actual uncertainty introduced by the item being calibrated, (e.g. resolution, repeatability) must also be considered and this, on its own, could result in the actual measurement uncertainty achievable on a customer's site being larger than the CMC.

[4] In the statement of CMC, L is the numerical value of the nominal length of the device measured in inches and R is the numerical value of the resolution of the device in microinches unless otherwise noted.

[5] The CMC stated for calibrations performed in the laboratory is applicable for calibrations performed in the field.

[6] CMC for the Wavetek 9100A is based on 1-year specifications within a temperature range of 23 °C ± 5 °C. Field calibrations will be performed within 23 °C ± 5 °C .

[7] CMC for the HP 34401 is based on 1-year specifications within a temperature range of 18 °C to 28 °C. Field calibrations will be performed within 18 °C to 28 °C, 30 % to 55 % humidity.

[8] Where ranges are not specified, the CMC stated is for the cardinal points only.

[9] In the statement of CMC, percentage (%) refers to percent of reading, unless otherwise noted.

[10] "SuperMic" is a registered trade mark with a last listed owner of Pratt & Whitney Measurement Systems, Inc., Connecticut U.S.A.

Accredited Laboratory

A2LA has accredited

ALDINGER COMPANY

Dallas, TX

for technical competence in the field of

Calibration

This laboratory is accredited in accordance with the recognized International Standard ISO/IEC 17025:2005 *General requirements for the competence of testing and calibration laboratories*. This laboratory also meets the requirements of ANSI/NCSLI Z540-1-1994 and any additional program requirements in the field of calibration. This accreditation demonstrates technical competence for a defined scope and the operation of a laboratory quality management system (*refer to joint ISO-ILAC-IAF Communiqué dated 8 January 2009*).

Presented this 19th day of May 2016.

Senior Director of Quality and Communications
For the Accreditation Council
Certificate Number 1509.01
Valid to June 30, 2018

For the calibrations to which this accreditation applies, please refer to the laboratory's Calibration Scope of Accreditation.